

Faisal Yusuf Ughratdar

(773) 302-7789 | faisalyusuf@gmail.com | linkedin.com/in/mechengineerfaisal | Chicago, IL

EDUCATION

University of Illinois Chicago

Bachelor of Science in Mechanical Engineering, GPA: 3.25/4.0

Chicago, IL

Expected May 2026

- Dean's List recipient

TECHNICAL SKILLS

CAD & Simulation: SolidWorks (motion simulation, assemblies), AutoCAD, Unreal Engine, ANSYS

Programming & Controls: Python, MATLAB, C++, Arduino, Raspberry Pi, ROS 2

Manufacturing: CNC machining, FDM 3D printing, DFM principles, GD&T, material selection, process optimization

Analysis: FEA, CFD, structural analysis, inverse kinematics, thermal analysis, statistical process control

ENGINEERING PROJECTS

Pentagon Robot

August 2025 – Present

University of Illinois Chicago, Department of Mechanical Engineering

Chicago, IL

- Leading mechanical design and controls integration of an autonomous five-bar parallel planar robot achieving cycle times under 5 seconds and placement accuracy within ± 0.5 mm
- Applied Model-Based Design (MBD) in SolidWorks for full mechanical assembly, motion simulation, and workspace validation to eliminate link interference and optimize kinematic performance
- Designed low-inertia closed-loop architecture using dual NEMA 17 stepper motors with real-time motor control via Raspberry Pi and Arduino UNO R3
- Integrated Raspberry Pi for inverse kinematics computation and path planning; performed structural analysis, material selection, and cost optimization for scalable production to 1,000 units within a \$350 prototype budget

PROFESSIONAL EXPERIENCE

Computer Specialist

August 2023 – Present

University of Illinois Chicago, College of Nursing

Chicago, IL

- Deployed and configured 200+ laptops and desktops using Microsoft Intune, ensuring compliance with security protocols
- Delivered Tier 1 and Tier 2 technical support for 100+ faculty and staff, resolving networking, VPN, and software issues with a 95% on-time resolution rate
- Managed help-desk ticketing, software updates, and patch management to minimize downtime and maintain service availability
- Implemented proactive maintenance workflows and system monitoring, reducing critical failures by 20%

Thermodynamics Laboratory Assistant

Fall 2024

University of Illinois Chicago, Department of Mechanical Engineering

Chicago, IL

- Prepared and supported experiments involving heat exchangers, combustion systems, and thermodynamic cycle analysis
- Guided students through experimental procedures, data collection, and uncertainty analysis
- Validated experimental results through statistical comparison with analytical models

LEADERSHIP & ACTIVITIES

Student Ambassador (Volunteer) – UIC Engineering Outreach

Fall 2024

- Led hands-on engineering demonstrations for 500+ prospective students and families during campus recruitment events
- Coordinated event logistics and volunteer teams to promote STEM outreach